C Communication

C.1 Introduction of communication protocol of the auto hematology analyzers

C.1.1 Messages supported by the HL7 interface protocol

The IPU software of the auto hematology analyzers and the LIS system enable the connection between the analyzer and the computer of the lab through the Ethernet. The analyzer could send the analysis results to the lab computer and receive the worklist information from it.

This communication protocol is defined based on the HL7 standard. HL7 is the digital data switching standard used in the medical field. It is firstly defined by America and now adopted by many countries. The definition is based on HL7 v2.3.1. For details of the HL7, please see HL7 Interface Standards Version 2.3.1.

C.1.2 Bottom transmitting layer protocol

The IPU software sends messages through TCP connection and the communication procedures consist of 3 phases:

Connecting

After starting up, the IPU software connects the LIS server actively according to the settings. If the connecting is failed, it retries; if the connecting is successful, it keeps the connection to make sure the data can be sent at any time. If the connection is found disconnected during operating, it retries to connect.

Data transmitting

Besides batch sending the data at the List Review and QC screen, if auto-communication is enabled, the IPU software will send the message while the new sample results are obtained.

Sending and receiving the message are synchronous both for batch communication and auto-communication. i.e. when every message is sent, it will wait for the confirmation. If the confirmation is received within 10s, then a complete message is sent and the next message will be sent; if the confirmation is not received within the 10s, then it is regarded that the sending is failed and it will skip to the sending of the next message.

The communication of QC data records is similar to that of the analysis results: send messages at the QC screen or QC History screen. Wait for the confirmation after sending each QC data. If the confirmation is received within 10s, then the message has been sent successfully; if the confirmation is not received within 10s, then it is regarded that the sending is failed and it will skip to the sending of the next message.

The bidirectional LIS inquiry communication is different from the processes mentioned above. The IPU software will send an inquiry (including the sample ID) every time it opens the

bidirectional LIS communication, saves worklists or before counting. The LIS will respond with a HL7 message based on the message it received, and then IPU will fill in the worklist or perform counting according to the response. If there is no response within 10s after the inquiry was sent, it is regarded that the inquiry is failed.

Disconnecting

When exiting the IPU software, the connection will be closed actively. When changing the communication settings, the connection will also be disconnected and then re-connect according to the new settings.

C.1.3 HL7 message layer protocol

HL7 top message protocol

The data of sample results etc. are transmitted in the form of UTF-8 coding strings.
The message strings are composed as per the HL7 standard. A message consists of several segments, each segment consists of several fields, a field consists of several components, and component consists of several sub components. The segment, field, component and sub component are divided by separators. The structure of the message is shown in Figure 1.

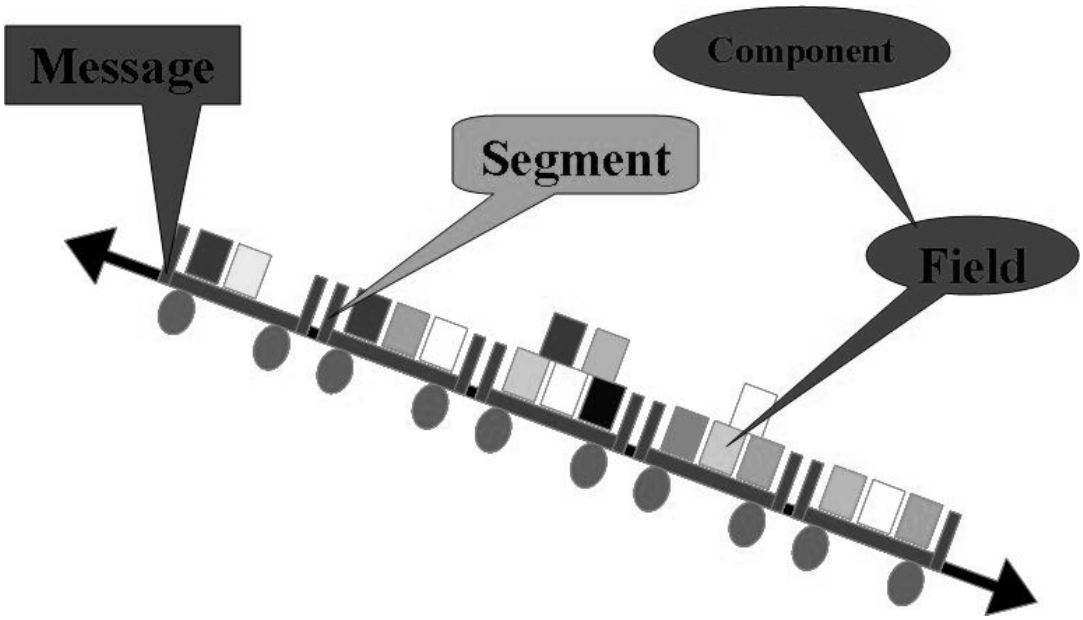

Figure 1 Structure of the message

A part of the HL7 message is shown below:

MSH|^~\&|BC-5380|Mindray|||20080617143943||ORU^R01|1|P|2.3.1||||||UNICODE

PID|1||7393670^^^^MR||Joan^Jlang||19900804000000|Female
PV1|1||nk^^001
OBR|1||20071207011|00001^Automated
Count^99MRC||20080508140600|20080508150616|||John||||20080508150000|||||||||HM||||||||
Mindray
OBX|1|IS|08001^Take Mode^99MRC||O||||||F
OBX|2|IS|08002^Blood Mode^99MRC||W||||||F
OBX|3|IS|08003^Test Mode^99MRC||CBC||||||F
OBX|4|IS|01002^Ref Group^99MRC||Woman||||||F
……

HL7 bottom protocol

TCP/IP is a protocol of byte stream. It doesn't provide the message boundary.HL7 of top protocol is based on messages. The function of terminating the message is not provided. In order to determine the message boundary, the bottom protocol of MLLP is used (such descriptions are also included in *HL7 Interface Standards Version 2.3.1.*).

Communication level

Messages are transmitted in the following format:

<SB> *ddddd* <EB><CR>

Among them:

<SB> = Start Block character (1 byte)

ASCII <VT>,i.e., <0x0B>.Do not confuse with the SOH or STX character in ASCII.

***ddddd* = Data (variable number of bytes)**

ddddd is the effective data of HL7 message and expressed in the form of string. For the strings used in the HL7 interface messages of auto hematology analyzers, the UTF-8 code is used.

<EB> = End Block character (1 byte)

ASCII <FS>,i.e. <0x1C>. Do not confuse with the ETX or EOT character in ASCII.

<CR> = Carriage Return (1 byte)

ASCII carriage return character, i.e. <0x0D>.

C.2 Introduction of HL7

C.2.1 HL7 basic grammar

Message constructing principles

Every HL7 message consists of several segments and ends up with the <CR> character.

Each segment consists of the segment name of three characters and field of changeable characters, and each field consists of the component and subcomponent. For each message, the separators of the field, component and subcomponent are defined in the MSH segment.

For example:

MSH|^~\&|BC-5380|Mindray|||20080617143943||ORU^R01|1|P|2.3.1||||||UNICODE

In this message:

The five characters following MSH define the separators to distinguish each field, component and subcomponent. Although they can be any non-text characters, HL7 standard recommends the characters in the table below:

Character	Meaning
|	Field separator
^	Component separator
&	Subcomponent separator
~	Repetition separator
\	ESC

The first field of MSH includes every separator. Some field behind are empty because they are optional and not used by Mindray HL7 interface. Detailed field definition and selection will be stated in the following contents.

For message of any type, the segments behind MSH appear in the fixed order. The order will be described in the following contents and the grammar is used to organize the segments order.

The segment appeared in [] is optional.

The segment appeared in {} can be repeated once or more.

String transferring principles

For the field data of ST, TX, FT, and CF, etc., separators may be contained in the string data like remark, clinical diagnosis and customized gender etc. When coding, the separators in the original strings shall be transferred into transferred character sequence; then, restore them when decoding. The transferring principles are shown in the table:

Transferred character	Original character
\F\	Field separator
\S\	Component separator
\T\	Subcomponent separator
\R\	Repetition separator
\E\	Transferred separator
\.br\	<CR>,i.e. end character of segment

Note: "\" in the transferred character sequence represents the transferred separator. Its value is defined in MSH segment.

C.2.2 HL7 data types

All the data information can be expressed by different types of HL7 fields. Only part of the HL7 standard is used in the communication protocol, see **D4 Appendix** for details.

C.3 Duplex communication

C.3.1 HL7 message supported

Process of duplex communication

1. The main unit directly sends the test results (or QC data) to LIS as Figure 2 shows.

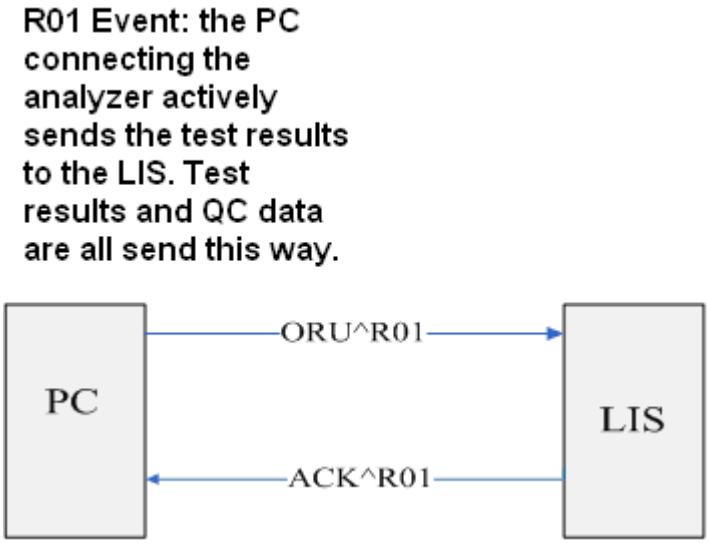

Figure 2 Test results (QC data) communication process

2. Worklist information searching

Worklist belongs to the Order message. Thus, the corresponding HL7 messages: ORM(General Order Message) and ORR(General Order Response Message) can be used. The communication process is shown in Figure 3.

Figure 3 Worklist searching communication process

Mostly used messages:

ORU^R01 message: it is mostly used for the transmission of the test results and QC data.

ORU Observational Results (Unsolicited) Description

MSHMessage header, necessary, including the communication information of message No., sending time, message separator and coding method, etc

{

PID Patient basic information, including patient name, gender, patient ID and birthday, etc

[PV1] Patient visit information, including patient type, department, bed No. and charge, etc

{

OBRsample information, including sample No., operator and run time, etc

{[OBX]} test data, including test results and work mode, etc

}

}

ACK^R01 message: it confirms the received ORU^R01 message.

ACK Acknowledgment Description

MSHMessage header

MSAMessage affirm, describing whether the communication message is received successfully

ORM^O01 message: Common order message, all the actions related to order basically use the message of this type. For example, create a new order or cancel an order. Here, the main unit requests LIS to re-fill the order message.

ORM General Order Message Description

MSH Message header

{ORC} Common message of Order, including the No. information of the sample searched

ORR^O02 message: affirming of the ORM^O01 message. Here, returning the completed information of order (i.e. worklist).

ORR^O02 General Order Response Message Description

MSH Message header

MSAMessage affirm

[PIDPatient basic information

[PV1]]Patient visit information

{

ORCCommon message of Order, including the sample No.

[

OBRsample information

{[OBX]}Data of other sample information, including work mode, etc.

]

}

C.3.2 HL7 segment definition involved

Detailed definition of fields contained in each segment will be listed in the table below. The meaning of each column is explained below.

1. No.: the HL7 message initiates with the segment name of 3 characters. The following each field will follow a separator, and the No. is the position order of the field.

For example:

PID |1 | |7393670^^^^MR||Joan^JIang||19900804000000|Female

↑ ↑ ↑

Segment name filed 1 filed 3

Note: the MSH message is a little different. The separator following the segment name is regarded as the first field and used to describe the value of the separators used in the message.

2. Field name: the logical meaning of the field

3. Data type: the HL7 standard type of the data, the structure will be described in Appendix A;

4. Recommended max length: the HL7 standard recommended length. But, during the actual transmitting, the length may exceed the length, so the separators should be identified to read the message when decoding the message.

5. Note: the note for the actual value of the fields

6. Samples: the sample of actual field value

MSH

The MSH(Message Header)segment contains basic information of HL7 message including separators' value, message type and coding method etc. It is the first field of every HL7 message.

Message used for example:

MSH|^~\&|BC-5380|Mindray|||20080617143943||ORU^R01|1|P|2.3.1||||||UNICODE

See Table 1 for definition of each field used in MSH segment.

Table 1 MSH field definitions

No.	Field Name	Data type	Recommended max length	Note	Samples
1	Field Separator	ST	1	Includes the separator of the first field after the segment name; be used to determine the separator's value of the rest parts of the message.	|
2	Encoding Characters	ST	4	Includes component separators, repetition separators, transferred separators and subcomponent separators; the value in the HL7 message of auto hematology analyzers is "^~\&"	^~\&
3	Sending application	EI	180	Application program of sending terminal. If the main unit sends the message; the value is "BC-5300" or "BC-5380".	BC-5380
4	Sending Facility	EI	180	Device of sending terminal. If the main unit sends the message, the value is "Mindray".	Mindray
7	Date/Time Of Message	TS	26	Created time of message (in the format of YYYY[MM[DD[HH[MM[SS]]]]]); adopts the system time.	20080617143943
9	Message Type	CM	7	Message type; in the format of "message type^event type". e.g. ORU^R01	ORU^R01
10	Message Control ID	ST	20	Message control ID; be used to mark a message uniquely.	1
11	Processing ID	PT	3	Message processing ID values: "P"- sample and worklist searching information; "D"- QC setup information; "T" – QC results information; In Ack messages, it is consistent with the previously	P

				received message.	
12	Version ID	VID	60	HL7 version information; the value is “2.3.1”.	2.3.1
18	Character Set	ID	10	Character set. The value is “UNICODE”, and the message is expressed by Unicode string.	UNICODE

MSA

The MSA(Message Acknowledgement) segment contains message confirming information.

Message used for example:

MSA|AA|1

See Table 2 for definition of the fields used.

Table 2 MSA field definitions

No.	Field Name	Data type	Recommended max length	Note	Samples
1	Acknowledgment Code	ID	2	Acknowledgement code: “AA”- receive, “AE” – error, “AR”- reject	AA
2	Message Control ID	ST	20	Message control ID; it’s consistent with the MSH-10 of the received message.	1
6	Error Condition	CE	100	Error condition (status code); it also contains error condition specification information; see Table 3 for the value.	

Table 3 Error code of MSA-6 field

Status code (MSA-6)	Status text (MSA-3)	Description/Remark
Successful:		AA
0	Message accepted	Successful
Error status code:		AE
100	Segment sequence error	Segment order in the message is wrong, or necessary segment lost
101	Required field missing	Necessary field lost in a segment

102	Data type error	Segment data type error, e.g. numbers are replaced by characters
103	Table value not found	Table value is not found; not used temporarily
Rejection status code:		AR
200	Unsupported message type	Message type is not supported
201	Unsupported event code	Event code is not supported
202	Unsupported processing id	Processing ID is not supported
203	Unsupported version id	Version ID is not supported
204	Unknown key identifier	Unknown key identifier, e.g. transmitting the patient information that is not exited
205	Duplicate key identifier	Repeated key words existed
206	Application record locked	Issues can not be executed at application program saving level, e.g. database is locked
207	Application internal error	Other interior errors of application program

PID

The PID(Patient Identification) segment contains the patient basic information.

Message used for example:

PID|1||7393670^^^^MR||Joan^JIang||19900804000000|Female

See Table 4 for definition of the fields used.

Table 4 PID field definitions

No.	Field Name	Data type	Recommended max length	Note	Samples
1	Set ID - PID	SI	4	Sequence NO.; it is used to mark the different PID segments of a message.	1
3	Patient Identifier List	CX	20	To be used as the patient ID in the message of the sample test results, in the form of “Patient ID^^^^MR”.	7393670^^^^MR

				To be used as QC lot No. in the message of QC.	
5	Patient Name	XPN	48	Patient name (dividing into two parts when sending: “FirstName” and “LastName”), e.g. “LastName^FirstName”.	Joan^JIang
7	Date/Time of Birth	TS	26	To be used as birthday in the message of sample results To be used as expiration date in the message of QC In the form of YYYY[MM[DD[HH[MM[SS]]]]]	19900804000000
8	Sex	IS	1	Gender, string.	Female

PV1

The PV1(Patient Visit) segment contains the patient visit information.

Message used for example:

PV1|1||nk^^001

See Table 5 for definition of the fields used.

Table 5 PV1 field definitions

No.	Field Name	Data type	Recommended max length	Note	Samples
1	Set ID - PV1	SI	4	Sequence NO.; it is used to mark the different PV1 segments of a message.	1
3	Assigned Patient Location	PL	80	Patient location information; in the form of “Department^ ^Bed No.”	nk^^001

OBR

The OBR(Observation Request) segment contains the test report information.

Message used for example:

OBR|1||20071207011|00001^Automated Count^99MRC||20080508140600|20080508150616|||John||||20080508150000||||||||||HM||||||||Mindray

See Table 6 for definition of the fields used.

Table 6 OBR field definitions

No.	Field Name	Data type	Recommended max length	Note	Samples
1	Set ID - OBR	SI	10	Sequence NO.; it is used to indicate the different OBR segments of a message.	1
2	Placer Order Number	EI	22	To be used as sample ID in the message of worklist searching response, i.e. ORC^O02	
3	Filler Order Number +	EI	22	To be used as sample ID in the message of test results To be used as file No. in the QC message	20071207011
4	Universal Service ID	CE	200	Universal service ID, to identify different types of test results. See Appendix B for detailed values.	00001^Automated Count^99MRC
6	Requested Date/time	TS	26	Requested Date/time To express the sampling date and time.	20080508140600
7	Observation Date/Time #	TS	26	Run Time	20080508150616
10	Collector Identifier *	XCN	60	Sample collector To indicate the deliverer	John
13	Relevant Clinical Info.	ST	300	Relevant clinical information. It can be used as	

				the clinical diagnostic information in the patient info.	
14	Specimen Received Date/Time *	TS	26	Sample received time To express the delivery time.	20080508150000
15	Specimen Source *	CM	300	Sample source Its value in HL7 message on the auto hematology analyzers: “BLDV”- Venous blood “BLDC”- Capillary blood	
22	Results Rpt/Status Chng - Date/Time +	TS	26	Results report/Status Change - Date/Time To be used as validating time.	
24	Diagnostic Serv Sect ID	ID	10	Diagnostic ID, the value is “HM”, means Hematology.	HM
28	Result Copies To	XCN	60	Result copies to To indicate the validater.	
32	Principal Result Interpreter +	CM	200	Principal result interpreter To be used as tester in the sample message To be used as “set by” in the QC message To be used as “Operator” in the QC run message	Mindray

OBX

The OBX(Observation/Result) segment contains the parameter information of each test result.
Message used for example:
OBX|6|NM|6690-2^WBC^LN||9.81|10*9/L|4.00-10.00|N|||F||E
See Table 7 for definition of the fields used.

Table 7 OBX field definitions

No.	Field Name	Data type	Recommended max length	Note	Samples
1	Set ID - OBX	SI	10	Sequence NO.; it is used to mark the different OBX segments of a message.	6
2	Value Type	ID	3	Data type of test results; the values can be “ST”, “NM”, “ED” and “IS”, etc.	NM
3	Observation Identifier	CE	590	Test item mark Form: “ID^Name^EncodeSys”. The “ID is the test item mark; “Name” the description information of the test item; “EncodeSys” is the coding system of the test item. For the values of the code of each test item, please see configuration file and Appendix B. Note: “ID” and “EncodeSys” are used to identify a unique parameter, but “Name” is used for description only.	6690-2^WBC^LN
5	Observation Value	*	65535	Test results data. It can be numbers, strings, enumeration values and binary data, etc., see Appendix B for their	9.81

				values (for the binary data, they are transferred by the Base64 coding method, see Appendix C for details).	
6	Units	CE	90	Units of test items. ISO standard units are used. The units used for communication are listed in Appendix B.	10*9/L
7	References Range	ST	90	Reference range; in the form of “lower limit-upper limit”, “< upper limit” or “> lower limit”.	4.00-10.00
8	Abnormal Flags	ID	5	Result flags: “N”- Normal “A”- Abnormal “H”- higher than upper limit “L”- lower than lower limit Note: The flag for normal or abnormal and that for high or low result may be displayed in this field at the same time. In this case, the two flags should be connected with a “~”, e.g.: “H~A”	N
11	Observ Result Status	ID	1	Test result status. The value is “F” - (Final Result);it means the final result.	F
13	User Defined Access Checks	ST	20	Customized contents. It stands for reagent expiration and modification mark, etc. The form is “mark 1-mark 2”. There are 3 kinds of marks in all:	E

				O – Expired reagent E – Active editing e – Passive editing	

ORC

The ORC(Common Order) segment contains the common information of order.

Message used for example:

ORC|RF||SampleID||IP

See Table 8 for definition of the fields used.

Table 8 ORC field definitions

No.	Field Name	Data type	Recommended max length	Note	Samples
1	Order Control	ID	2	Order control word In the ORM message the value is “RF” which means “re-fill the order request”. In the ORR message the value is “AF” which means “affirm the re-filled order”.	RF
2	Placer Order Number	EI	22	Placer order number In the ORM message the value is empty; in the ORR message the value is the sample ID.	
3	Filler OrderNum	EI	22	Filler Order Number In the ORM message the value is the sample ID; in the ORR message the value is empty.	SampleID
5	Order Status	ID	2	Order status In the ORM message the value is “IP” which means “order is being processed, but results are not obtained”; in the ORR message the value is empty.	IP

C.3.3 Example of a complete message

The following two messages demonstrate the communication process of the sample data.

Sample message

MSH|^~\&|BC-5380|Mindray|||20080617143943||ORU^R01|1|P|2.3.1||||||UNICODE
PID|1||7393670^^^^MR||Joan^JIang||19900804000000|Female
PV1|1||nk^^001
OBR|1||20071207011|00001^Automated
Count^99MRC||20080508140600|20080508150616|||John||||20080508150000||||||||||HM||||||||
Mindray
OBX|1|IS|08001^Take Mode^99MRC||O||||||F
OBX|2|IS|08002^Blood Mode^99MRC||W||||||F
OBX|3|IS|08003^Test Mode^99MRC||CBC||||||F
OBX|4|IS|01002^Ref Group^99MRC||Woman||||||F
OBX|5|NM|30525-0^Age^LN||18|yr|||||F
OBX|6|NM|6690-2^WBC^LN||9.81|10*9/L|4.00-10.00|N|||F||E
OBX|7|NM|704-7^BAS#^LN|||10*9/L|0.00-0.10||||F
OBX|8|NM|706-2^BAS%^LN||||0.000-0.010||||F
OBX|9|NM|751-8^NEU#^LN|||10*9/L|2.00-7.00||||F
OBX|10|NM|770-8^NEU%^LN||||0.500-0.700||||F
OBX|11|NM|711-2^EOS#^LN|||10*9/L|0.02-0.50||||F
OBX|12|NM|713-8^EOS%^LN||||0.005-0.050||||F
OBX|13|NM|731-0^LYM#^LN|||10*9/L|0.80-4.00||||F
OBX|14|NM|736-9^LYM%^LN||||0.200-0.400||||F
OBX|15|NM|742-7^MON#^LN|||10*9/L|0.12-0.80||||F
OBX|16|NM|5905-5^MON%^LN||||0.030-0.080||||F
OBX|17|NM|26477-0^*ALY#^LN|||10*9/L|0.00-0.20||||F
OBX|18|NM|13046-8^*ALY%^LN||||0.000-0.020||||F
OBX|19|NM|10000^*LIC#^99MRC|||10*9/L|0.00-0.20||||F
OBX|20|NM|10001^*LIC%^99MRC||||0.000-0.025||||F
OBX|21|NM|789-8^RBC^LN||4.53|10*12/L|3.50-5.00|N|||F
OBX|22|NM|718-7^HGB^LN||65|g/L|110-150|L|||F
OBX|23|NM|787-2^MCV^LN||89.5|fL|80.0-100.0|N|||F
OBX|24|NM|785-6^MCH^LN||14.4|pg|27.0-31.0|L|||F
OBX|25|NM|786-4^MCHC^LN||160|g/L|320-360|L|||F
OBX|26|NM|788-0^RDW-CV^LN||0.133||0.115-0.145|N|||F
OBX|27|NM|21000-5^RDW-SD^LN||50.9|fL|35.0-56.0|N|||F
OBX|28|NM|4544-3^HCT^LN||0.405||0.370-0.480|N|||F
OBX|29|NM|777-3^PLT^LN||212|10*9/L|100-300|N|||F
OBX|30|NM|32623-1^MPV^LN||6.6|fL|7.0-11.0|L|||F
OBX|31|NM|32207-3^PDW^LN||15.4||15.0-17.0|N|||F
OBX|32|NM|10002^PCT^99MRC||1.40|mL/L|1.08-2.82|N|||F
OBX|33|IS|12014^Anemia^99MRC||T||||||F

OBX|34|IS|15180-3^Hypochromia^LN||T||||||F

OBX|35|NM|15001^WBC Histogram. Left Line^99MRC||7||||||F

OBX|36|NM|15002^WBC Histogram. Right Line^99MRC||65||||||F

OBX|37|NM|15003^WBC Histogram. Middle Line^99MRC||30||||||F

OBX|38|ED|15008^WBC Histogram. BMP^99MRC||^Image^BMP^Base64^ … … WBC Histogram bmp data……||||||F

OBX|39|NM|15051^RBC Histogram. Left Line^99MRC||26||||||F

OBX|40|NM|15052^RBC Histogram. Right Line^99MRC||164||||||F

OBX|41|ED|15056^RBC Histogram. BMP^99MRC||^Image^BMP^Base64^ … … RBC Histogram bmp data……||||||F

OBX|42|NM|15111^PLT Histogram. Left Line^99MRC||3||||||F

OBX|43|NM|15112^PLT Histogram. Right Line^99MRC||43||||||F

OBX|44|ED|15116^PLT Histogram. BMP^99MRC||^Image^BMP^Base64^……PLT Histogram bmp data……||||||F

OBX|45|ED|15200^WBC DIFF Scattergram. BMP^99MRC||^Image^BMP^Base64^……WBC Diff Scattergram bmp data……||||||F

OBR|2||20071207011|00002^Manual Count^99MRC||||||||||BLDV

OBX|46|NM|747-6^Myeloblasts%. Manual^LN||0.0|%|||||F

OBX|47|NM|783-1^Promyelocytes%. Manual^LN||0.0|%|||||F

OBX|48|NM|749-2^Myelocytes%. Manual^LN||0.0|%|||||F

OBX|49|NM|740-1^Metamyelocyte%. Manual^LN||0.0|%|||||F

OBX|50|NM|764-1^Neuts Band%. Manual^LN||0.0|%|||||F

OBX|51|NM|769-0^Neuts Seg%. Manual^LN||0.0|%|||||F

OBX|52|NM|714-6^Eosinophils%. Manual^LN||0.0|%|||||F

OBX|53|NM|707-0^Basophils%. Manual^LN||0.0|%|||||F

OBX|54|NM|33831-9^Lymphoblasts%. Manual^LN||0.0|%|||||F

OBX|55|NM|6746-2^Prolymphocytes%. Manual^LN||0.0|%|||||F

OBX|56|NM|737-7^Lymphocytes%. Manual^LN||0.0|%|||||F

OBX|57|NM|29261-5^Abnormal Lymphs%. Manual^LN||0.0|%|||||F

OBX|58|NM|33840-0^Monoblasts%. Manual^LN||0.0|%|||||F

OBX|59|NM|13599-6^Promonocytes%. Manual^LN||0.0|%|||||F

OBX|60|NM|744-3^Monocytes%. Manual^LN||0.0|%|||||F

OBX|61|NM|18309-5^NRBCs%. Manual^LN||0.0|%|||||F

OBX|62|NM|31112-6^Reticulocytes%. Manual^LN||0.0|%|||||F

OBX|63|NM|11000^Undefined Cells%. Manual^99MRC||0.0|%|||||F

OBX|64|NM|11001^Other Abnormal Cells%. Manual^99MRC||0.0|%|||||F

Sample response message

Every time a sample result is received, a sample response message composed of two

message segments (MSH and MSA) will be sent. To send a correct response message, take into consideration that: the MSH-9 field should be ACK^R01 which indicates that it is a sample response message; If the value in the MSA-2 field is the same with the MSH-10 value of the analysis result, it indicates that this response message is corresponding to the sent analysis result. The MSA-2 value in the following example is 1

MSH|^~\&|LIS||||20080617143944||ACK^R01|1|P|2.3.1||||||UNICODE
MSA|AA|1

QC message

The content of the QC message differs from that of the sample analysis result: the MSH-11 value of the QC message is Q which indicates that it is a QC message; each QC message is corresponding to one QC point in the IPU software which may contain several analysis results. For example, there is one analysis result in an L-J QC message, while there are two analysis results and one mean calculation result in an X-R QC message.
A QC messasge is composed of an MSH message head and several analysis results, each of which contains the PID and OBR segments as the head of the sample message, as well as several OBX segments to carry parameters and other information. The OBR-4 field of each analysis result indicates the type of the result (X-R analysis result, X-R mean or L-J analysis result). See Appendix: Message coding definition for details.
An example of the X-R QC message is shown as follows:

MSH|^~\&|BC-5380|Mindray|||20081120171602||ORU^R01|1|Q|2.3.1||||||UNICODE
PID|1||6666666||||20080807235959
OBR|1||6|00006^XR QCR^99MRC|||20080807142518||||||||||||||||HM|||||||R&D Engineer
OBX|1|IS|05001^Qc Level^99MRC||M||||||F
OBX|2|IS|08001^Take Mode^99MRC||C||||||F
OBX|3|IS|08002^Blood Mode^99MRC||W||||||F
OBX|4|NM|6690-2^WBC^LN||0.00|10*9/L|||||F
OBX|5|NM|704-7^BAS#^LN||***.**|10*9/L|||||F
OBX|6|NM|706-2^BAS%^LN||**.*|%|||||F
OBX|7|NM|751-8^NEU#^LN||***.**|10*9/L|||||F
OBX|8|NM|770-8^NEU%^LN||**.*|%|||||F
OBX|9|NM|711-2^EOS#^LN||***.**|10*9/L|||||F
OBX|10|NM|713-8^EOS%^LN||**.*|%|||||F
OBX|11|NM|731-0^LYM#^LN||***.**|10*9/L|||||F
OBX|12|NM|736-9^LYM%^LN||**.*|%|||||F
OBX|13|NM|742-7^MON#^LN||***.**|10*9/L|||||F
OBX|14|NM|5905-5^MON%^LN||**.*|%|||||F
OBX|15|NM|789-8^RBC^LN||0.02|10*12/L|||||F

OBX|16|NM|718-7^HGB^LN||0|g/L|||||F
OBX|17|NM|787-2^MCV^LN||***.*|fL|||||F
OBX|18|NM|785-6^MCH^LN||***.*|pg|||||F
OBX|19|NM|786-4^MCHC^LN||****|g/L|||||F
OBX|20|NM|788-0^RDW-CV^LN||**.*|%|||||F
OBX|21|NM|21000-5^RDW-SD^LN||***.*|fL|||||F
OBX|22|NM|4544-3^HCT^LN||0.0|%|||||F
OBX|23|NM|777-3^PLT^LN||4|10*9/L|||||F
OBX|24|NM|32623-1^MPV^LN||**.*|fL|||||F
OBX|25|NM|32207-3^PDW^LN||**.*||||||F
OBX|26|NM|10002^PCT^99MRC||.***|%|||||F
OBX|27|NM|10003^GRAN-X^99MRC||6||||||F
OBX|28|NM|10004^GRAN-Y^99MRC||32||||||F
OBX|29|NM|10005^GRAN-Y(W)^99MRC||20||||||F
OBX|30|NM|10006^WBC-MCV^99MRC||83||||||F
OBX|31|NM|15001^WBC Histogram. Left Line^99MRC||10||||||F
OBX|32|NM|15002^WBC Histogram. Right Line^99MRC||72||||||F
OBX|33|NM|15003^WBC Histogram. Middle Line^99MRC||37||||||F
OBX|34|ED|15008^WBC Histogram. BMP^99MRC||^Image^BMP^Base64^……WBC histogram data……||||||F
OBX|35|NM|15051^RBC Histogram. Left Line^99MRC||10||||||F
OBX|36|NM|15052^RBC Histogram. Right Line^99MRC||250||||||F
OBX|37|ED|15056^RBC Histogram. BMP^99MRC||^Image^BMP^Base64^……RBC histogram data……||||||F
OBX|38|NM|15111^PLT Histogram. Left Line^99MRC||3||||||F
OBX|39|NM|15112^PLT Histogram. Right Line^99MRC||24||||||F
OBX|40|ED|15116^PLT Histogram. BMP^99MRC||^Image^BMP^Base64^……PLT histogram data……||||||F
OBX|41|ED|15200^WBC DIFF Scattergram. BMP^99MRC||^Image^BMP^Base64^……DIFF scattergram data……||||||F
PID|2||6666666||||20080807235959
OBR|2||6|00006^XR QCR^99MRC|||20080807142640||||||||||||||||HM|||||||R&D Engineer
OBX|42|IS|05001^Qc Level^99MRC||M||||||F
OBX|43|IS|08001^Take Mode^99MRC||C||||||F
OBX|44|IS|08002^Blood Mode^99MRC||W||||||F
OBX|45|NM|6690-2^WBC^LN||0.00|10*9/L|||||F
OBX|46|NM|704-7^BAS#^LN||***.**|10*9/L|||||F
OBX|47|NM|706-2^BAS%^LN||**.*|%|||||F
OBX|48|NM|751-8^NEU#^LN||***.**|10*9/L|||||F
OBX|49|NM|770-8^NEU%^LN||**.*|%|||||F

OBX|50|NM|711-2^EOS#^LN||***.**|10*9/L|||||F
OBX|51|NM|713-8^EOS%^LN||**.*|%|||||F
OBX|52|NM|731-0^LYM#^LN||***.**|10*9/L|||||F
OBX|53|NM|736-9^LYM%^LN||**.*|%|||||F
OBX|54|NM|742-7^MON#^LN||***.**|10*9/L|||||F
OBX|55|NM|5905-5^MON%^LN||**.*|%|||||F
OBX|56|NM|789-8^RBC^LN||0.02|10*12/L|||||F
OBX|57|NM|718-7^HGB^LN||0|g/L|||||F
OBX|58|NM|787-2^MCV^LN||***.*|fL|||||F
OBX|59|NM|785-6^MCH^LN||***.*|pg|||||F
OBX|60|NM|786-4^MCHC^LN||****|g/L|||||F
OBX|61|NM|788-0^RDW-CV^LN||**.*|%|||||F
OBX|62|NM|21000-5^RDW-SD^LN||***.*|fL|||||F
OBX|63|NM|4544-3^HCT^LN||0.0|%|||||F
OBX|64|NM|777-3^PLT^LN||5|10*9/L|||||F
OBX|65|NM|32623-1^MPV^LN||**.*|fL|||||F
OBX|66|NM|32207-3^PDW^LN||**.*||||||F
OBX|67|NM|10002^PCT^99MRC||.***|%|||||F
OBX|68|NM|10003^GRAN-X^99MRC||28||||||F
OBX|69|NM|10004^GRAN-Y^99MRC||19||||||F
OBX|70|NM|10005^GRAN-Y(W)^99MRC||20||||||F
OBX|71|NM|10006^WBC-MCV^99MRC||60||||||F
OBX|72|NM|15001^WBC Histogram. Left Line^99MRC||10||||||F
OBX|73|NM|15002^WBC Histogram. Right Line^99MRC||72||||||F
OBX|74|NM|15003^WBC Histogram. Middle Line^99MRC||37||||||F
OBX|75|ED|15008^WBC Histogram. BMP^99MRC||^Image^BMP^Base64^……WBC histogram data……||||||F
OBX|76|NM|15051^RBC Histogram. Left Line^99MRC||10||||||F
OBX|77|NM|15052^RBC Histogram. Right Line^99MRC||250||||||F
OBX|78|ED|15056^RBC Histogram. BMP^99MRC||^Image^BMP^Base64^ ……RBC histogram data……||||||F
OBX|79|NM|15111^PLT Histogram. Left Line^99MRC||3||||||F
OBX|80|NM|15112^PLT Histogram. Right Line^99MRC||24||||||F
OBX|81|ED|15116^PLT Histogram. BMP^99MRC||^Image^BMP^Base64^ ……PLT histogram data……||||||F
OBX|82|ED|15200^WBC DIFF Scattergram. BMP^99MRC||^Image^BMP^Base64^ ……DIFF scattergram data……||||||F
PID|3||6666666
OBR|3||6|00008^XR QCR Mean^99MRC||||||||||||||||||HM
OBX|83|NM|6690-2^WBC^LN||0.00|10*9/L|||||F

OBX|84|NM|704-7^BAS#^LN||***.**|10*9/L|||||F
OBX|85|NM|706-2^BAS%^LN||**.*|%|||||F
OBX|86|NM|751-8^NEU#^LN||***.**|10*9/L|||||F
OBX|87|NM|770-8^NEU%^LN||**.*|%|||||F
OBX|88|NM|711-2^EOS#^LN||***.**|10*9/L|||||F
OBX|89|NM|713-8^EOS%^LN||**.*|%|||||F
OBX|90|NM|731-0^LYM#^LN||***.**|10*9/L|||||F
OBX|91|NM|736-9^LYM%^LN||**.*|%|||||F
OBX|92|NM|742-7^MON#^LN||***.**|10*9/L|||||F
OBX|93|NM|5905-5^MON%^LN||**.*|%|||||F
OBX|94|NM|789-8^RBC^LN||0.02|10*12/L|||||F
OBX|95|NM|718-7^HGB^LN||0|g/L|||||F
OBX|96|NM|787-2^MCV^LN||***.*|fL|||||F
OBX|97|NM|785-6^MCH^LN||***.*|pg|||||F
OBX|98|NM|786-4^MCHC^LN||****|g/L|||||F
OBX|99|NM|788-0^RDW-CV^LN||**.*|%|||||F
OBX|100|NM|21000-5^RDW-SD^LN||***.*|fL|||||F
OBX|101|NM|4544-3^HCT^LN||0.0|%|||||F
OBX|102|NM|777-3^PLT^LN||5|10*9/L|||||F
OBX|103|NM|32623-1^MPV^LN||**.*|fL|||||F
OBX|104|NM|32207-3^PDW^LN||**.*||||||F
OBX|105|NM|10002^PCT^99MRC||.***|%|||||F
OBX|106|NM|10003^GRAN-X^99MRC||17||||||F
OBX|107|NM|10004^GRAN-Y^99MRC||26||||||F
OBX|108|NM|10005^GRAN-Y(W)^99MRC||20||||||F
OBX|109|NM|10006^WBC-MCV^99MRC||72||||||F

QC response message

The only difference between the QC response message and the analysis result response message is that the MSH-11 value of the QC response message is Q.
An example of the ACK X-R QC message is shown as follows:

MSH|^~\&|LIS|||20081120171602||ACK^R01|1|Q|2.3.1||||||UNICODE
MSA|AA|1

Bidirectional LIS inquiry message

A bidirectional LIS inquiry message contains a sample ID. After the LIS received the inquiry message, it will search for the corresponding patient and sample information to provide a response.

The inquiry message is composed of two message segments: MSH and ORC. The MSH segment is almost the same with that of the analysis result, except that the MSH-9 value is ORM^O01. The ORC-3 field should be filled with the receiver code (in this case, the sample ID; where in the following sample, it is SampleID1). Note that in the autoloading analysis, if there is a barcode scan error while sending an inquiry message, the sample ID will be “Invalid”.
An example of the inquiry message is shown as follows:

MSH|^~\&|BC-5380|Mindray|||20081120174836||ORM^O01|4|P|2.3.1||||||UNICODE
ORC|RF||SampleID1||IP

Bidirectional LIS inquiry response message

When the LIS received an inquiry message, it needs to send back an inquiry response message. The first two message segments of the inquiry response message are MSH and MSA. The MSH-9 field (indicating the type of the segment) is filled with ORR^O02, while the MSA segment should be filled up as shown in the following example of the inquiry response message. If the LIS gets searching results for the inquiry, there will be PID, PV1, ORC, OBR and OBX message segments after the two heading segments to provide the patient and sample information, in the same way as the sample data message does. The ORC segment is indispensable for an inquiry response message with searching results, in which the ORC-1 value is AF, and ORC-2 is the filter (the sample ID). Note that the OBR-2 field indicates the sample ID, which should be the same value as in the ORC-2 field; otherwise, the message will be regarded as incorrect.
An example of the inquiry response message with searching results is shown as follows:

MSH|^~\&|LIS||||20081120174836||ORR^O02|1|P|2.3.1||||||UNICODE
MSA|AA|4
PID|1||ChartNo^^^^MR||^FName||19810506|NT
PV1|1|nk^^Bn4||||||||||||||||NewCharge
ORC|AF|SampleID1|||
OBR|1|SampleID1||||20060506||||tester|||Diagnose
content....|20060504||||||||20080821||HM||||Validator||||Operator
OBX|1|IS|08001^Take Mode^99MRC||A||||||F
OBX|2|IS|08002^Blood Mode^99MRC||W||||||F
OBX|3|IS|08003^Test Mode^99MRC||CBC||||||F
OBX|4|IS|01002^Ref Group^99MRC||XXXX||||||F
OBX|5|NM|30525-0^Age^LN||1|hr|||||F
OBX|6|ST|01001^Remark^99MRC||remark content....||||||F

An example of the inquiry response message with no search result is shown as follows, in which the MSA-2 field indicates the result of the response. In this example, the MSA-2 value is

“AR”, indicating the inquiry was rejected; if it is “AE", then there is an error in the inquiry process.

MSH|^~\&|LIS||||20081120175238||ORR^O02|1|P|2.3.1|||||UNICODE
MSA|AR|9

C.4 Appendix: Definition of the HL7 data type used

- **CE - Code Element**

<identifier (ST)> ^ <text (ST)> ^ <name of coding system (ST)> ^ <alternate identifier (ST)> ^ <alternate text (ST)> ^ <name of alternate coding system (ST)>

- **CM - Composite**

The format is defined by the specific field.

- **CX - Extended composite ID with check digit**

<ID (ST)> ^ <check digit (ST)> ^ <code identifying the check digit scheme employed (ID)> ^ < assigning authority (HD)> ^ <identifier type code (IS)> ^ < assigning facility (HD)>

- **ED – Encapsulate Data**

<source application(HD)> ^ <type of data(ID)> ^ <data sub type(ID)> ^ <encoding(ID)> ^ <data(ST)>

- **EI - Entity Identifier**

<entity identifier (ST)> ^ <namespace ID (IS)> ^ <universal ID (ST)> ^ <universal ID type (ID)>

- **FC – Financial Class**

<financial class(IS)> ^ <effective date(TS)>

- **HD - Hierarchic designator**

<namespace ID (IS)> ^ <universal ID (ST)> ^ <universal ID type (ID)>
Used only as part of EI and other data types.

- **FT - Formatted text**

This data type is derived from the string data type by allowing the addition of embedded formatting instructions. These instructions are limited to those that are intrinsic and independent of the circumstances under which the field is being used.

- **IS - Coded value for user-defined tables**

The value of such a field follows the formatting rules for an ST field except that it is drawn from a site-defined (or user-defined) table of legal values. There shall be an HL7 table number associated with IS data types.

- **ID - Coded values for HL7 tables**

The value of such a field follows the formatting rules for an ST field except that it is drawn from a table of legal values. There shall be an HL7 table number associated with ID data types.

- **NM - Numeric**

A number represented as a series of ASCII numeric characters consisting of an optional leading sign (+ or -), the digits and an optional decimal point.

- **PL - Person location**

<point of care (IS)> ^ <room (IS)> ^ <bed (IS)> ^ <facility (HD)> ^ < location status (IS)> ^ <person location type (IS)> ^ <building (IS)> ^ <floor (IS)> ^ <location description (ST)>

- **PT - Processing type**

<processing ID (ID)> ^ <processing mode (ID)>

- **SI - Sequence ID**

A non-negative integer in the form of an NM field. The uses of this data type are defined in the chapters defining the segments and messages in which it appears.

- **ST – String**

- **TS - Time stamp**

YYYY[MM[DD[HHMM[SS[.S[S[S[S]]]]]]]][+/-ZZZZ] ^ <degree of precision>

- **XCN - Extended composite ID number and name**

In Version 2.3, use instead of the CN data type. <ID number (ST)> ^ <family name (ST)> & <last_name_prefix (ST) ^ <given name (ST)> ^ <middle initial or name (ST)> ^ <suffix (e.g., JR or III) (ST)> ^ <prefix (e.g., DR) (ST)> ^ <degree (e.g., MD) (ST)> ^ <source table (IS)> ^ <assigning authority (HD)> ^ <name type code (ID)> ^ <identifier check digit (ST)> ^ <code identifying the check digit scheme employed (ID)> ^ <identifier type code (IS)> ^ <assigning facility (HD)> ^ <name representation code (ID)>

- **XPN - Extended person name**

In Version 2.3, replaces the PN data type. <family name (ST)> ^ <given name (ST)> & <last_name_prefix (ST)> ^ <middle initial or name (ST)> ^ <suffix (e.g., JR or III) (ST)> ^ <prefix (e.g., DR) (ST)> ^ <degree (e.g., MD) (IS)> ^ <name type code (ID) > ^ <name representation code (ID)>

- **VID - Version identifier**

<version ID (ID)> ^ <internationalization code (CE)> ^ <international version ID (CE)>

C.5 Appendix: Message coding definition

1. In the HL7 message, the OBR-4(Universal Serview ID) field is used for identifying the type

of the test results, for example, to identify the results as sample results, microscope exam results or QC results in the form of “ID^Name^EncodeSys”. The coding values of the field are listed in the table below.

Table 9 OBR-4 Coding

Data	Coding (ID)	Name	EncodeSys
Analysis result	00001	Automated Count	99MRC
Microscope exam result	00002	Manual Count	99MRC
LJ QC result	00003	LJ QCR	99MRC
X QC result	00004	X QCR	99MRC
XB QC result	00005	XB QCR	99MRC
XR QC result	00006	XR QCR	99MRC
X QC result mean	00007	X QCR Mean	99MRC
XR QC result mean	00008	XR QCR Mean	99MRC

2. Each OBX segment contains one test parameter or information of other data and consists of the following fields: OBX-2, it indicates the HL7 type of the data contained; OBX-3, it is the mark of the data in the form of “ID^Name^EncodeSys”; OBX-5, it contains the value of the data; OBX-6, it contains the unit for the parameter, expressing in the ISO standard.

The HL7 types and coding marks of all the communication data are listed in Table 10. The units of all the communication data are listed in table 11.

Table 10 HL7 types and coding marks

Data	HL7 Type (OBX-2)	Coding (ID)	Name	EncodeSys	OBX-3 field example
Other data					
Take Mode	IS	08001	Take Mode	99MRC	08001^Take Mode^99MRC
Blood Mode	IS	08002	Blood Mode	99MRC	08002^Blood Mode^99MRC
Test mode	IS	08003	Test Mode	99MRC	08003^Test Mode^99MRC
Age	NM	30525-0	Age	LN	30525-0^Age^LN
Remark	ST	01001	Remark	99MRC	01001^Remark^99MRC
Ref Group	IS	01002	Ref Group	99MRC	01002^Ref Group^99MRC

QC Level	IS	05001	Qc Level	99MRC	05001^Qc Level^99MRC
Analysis results data					
WBC	NM	6690-2	WBC	LN	6690-2^WBC^LN
BAS	NM	704-7	BAS#	LN	704-7^BAS#^LN
BAS_PER	NM	706-2	BAS%	LN	706-2^BAS%^LN
NEU	NM	751-8	NEU#	LN	751-8^NEU#^LN
NEU_PER	NM	770-8	NEU%	LN	770-8^NEU%^LN
EOS	NM	711-2	EOS#	LN	711-2^EOS#^LN
EOS_PER	NM	713-8	EOS%	LN	713-8^EOS%^LN
LYM	NM	731-0	LYM#	LN	731-0^LYM#^LN
LYM_PER	NM	736-9	LYM%	LN	736-9^LYM%^LN
MON	NM	742-7	MON#	LN	742-7^MON#^LN
MON_PER	NM	5905-5	MON%	LN	5905-5^MON%^LN
ALY	NM	26477-0	*ALY#	LN	26477-0^*ALY#^LN
ALY_PER	NM	13046-8	*ALY%	LN	13046-8^*ALY%^LN
LIC	NM	10000	*LIC#	99MRC	10000^*LIC#^99MRC
LIC_PER	NM	10001	*LIC%	99MRC	10001^*LIC%^99MRC
RBC	NM	789-8	RBC	LN	789-8^RBC^LN
HGB	NM	718-7	HGB	LN	718-7^HGB^LN
MCV	NM	787-2	MCV	LN	787-2^MCV^LN
MCH	NM	785-6	MCH	LN	785-6^MCH^LN
MCHC	NM	786-4	MCHC	LN	786-4^MCHC^LN
RDW_CV	NM	788-0	RDW-CV	LN	788-0^RDW-CV^LN
RDW_SD	NM	21000-5	RDW-SD	LN	21000-5^RDW-SD^LN
HCT	NM	4544-3	HCT	LN	4544-3^HCT^LN
PLT	NM	777-3	PLT	LN	777-3^PLT^LN
MPV	NM	32623-1	MPV	LN	32623-1^MPV^LN
PDW	NM	32207-3	PDW	LN	32207-3^PDW^LN
PCT	NM	10002	PCT	99MRC	10002^PCT^99MRC
GRAN-X	NM	10003	GRAN-X	99MRC	10003^GRAN-X^99MRC
GRAN-Y	NM	10004	GRAN-Y	99MRC	10004^GRAN-Y^99M

					RC
GRAN-Y(W)	NM	10005	GRAN-Y(W)	99MRC	10005^GRAN-Y(W)^99MRC
WBCMCV	NM	10006	WBC-MCV	99MRC	10006^WBC-MCV^99MRC
Microscope exam data					
Blood Type	ST	882-1	Blood Type	LN	882-1^Blood Type^LN
WBC Morphology	ST	11156-7	WBC Morphology	LN	11156-7^WBC Morphology^LN
RBC Morphology	ST	6742-1	RBC Morphology	LN	6742-1^RBC Morphology^LN
PLT Morphology	ST	11125-2	PLT Morphology	LN	11125-2^PLT Morphology^LN
Myeloblast	NM	747-6	Myeloblasts%. Manual	LN	747-6^Myeloblasts%. Manual^LN
Promyelocyte	NM	783-1	Promyelocytes%. Manual	LN	783-1^Promyelocytes%. Manual^LN
Myelocyte	NM	749-2	Myelocytes%. Manual	LN	749-2^Myelocytes%. Manual^LN
MetaMyelocyte	NM	740-1	Metamyelocyte%. Manual	LN	740-1^Metamyelocyte%. Manual^LN
BandFormNeut	NM	764-1	Neuts Band%. Manual	LN	764-1^Neuts Band%. Manual^LN
SegmentNeut	NM	769-0	Neuts Seg%. Manual	LN	769-0^Neuts Seg%. Manual^LN
Eosinophils	NM	714-6	Eosinophils%. Manual	LN	714-6^Eosinophils%. Manual^LN
Basophils	NM	707-0	Basophils%. Manual	LN	707-0^Basophils%. Manual^LN
Lymphoblast	NM	33831-9	Lymphoblasts%. Manual	LN	33831-9^Lymphoblasts%. Manual^LN
Prolymphocytes	NM	6746-2	Prolymphocytes%. Manual	LN	6746-2^Prolymphocytes%. Manual^LN
Lymphocytes	NM	737-7	Lymphocytes%. Manual	LN	737-7^Lymphocytes%. Manual^LN
AbnLymph	NM	29261-5	Abnormal Lymphs%. Manual	LN	29261-5^Abnormal Lymphs%. Manual^LN

Monoblast	NM	33840-0	Monoblasts%. Manual	LN	33840-0^Monoblasts %. Manual^LN
Promonocytes	NM	13599-6	Promonocytes%. Manual	LN	13599-6^Promonocytes%. Manual^LN
Monocyte	NM	744-3	Monocytes%. Manual	LN	744-3^Monocytes%. Manual^LN
NRBCS	NM	18309-5	NRBCs%. Manual	LN	18309-5^NRBCs%. Manual^LN
Reticulocyte	NM	31112-6	Reticulocytes%. Manual	LN	31112-6^Reticulocytes%. Manual^LN
UndefinedCells	NM	11000	Undefined Cells%. Manual	99MRC	11000^Undefined Cells%. Manual^99MRC
OtherAbnormalCells	NM	11001	Other Abnormal Cells%. Manual	99MRC	11001^Other Abnormal Cells%. Manual^99MRC
ESR	NM	30341-2	ESR	LN	
Analysis results medium data(WBC, RBC, PLT histogram and scattergram data, etc.)					
WBC Histogram Binary Data	ED	15000	WBC Histogram. Binary	99MRC	15000^WBC Histogram Binaray^99MRC
WBC Histogram Left Discriminator	NM	15001	WBC Histogram. Left Line	99MRC	15001^WBC Histogram. Left Line^99MRC
WBC Histogram Right Discriminator	NM	15002	WBC Histogram. Right Line	99MRC	15002^WBC Histogram. Right Line^99MRC
WBC Histogram Middle Discriminator	NM	15003	WBC Histogram. Middle Line	99MRC	15003^WBC Histogram. Middle Line^99MRC
WBC Histogram Original Data Length	NM	15004	WBC Histogram. Meta Length	99MRC	15004^WBC Histogram. Meta Length^99MRC
WBC Histogram Left Discriminator Adjusted Mark	IS	15005	WBC Histogram. Left Line Adjusted	99MRC	15005^WBC Histogram. Left Line Adjusted^99MRC
WBC Histogram	IS	15006	WBC Histogram.	99MRC	15006^WBC

Right Discriminator Adjusted Mark			Right Line Adjusted		Histogram. Right Line Adjusted^99MRC
WBC Histogram Middle Discriminator Adjusted Mark	IS	15007	WBC Histogram. Middle Line Adjusted	99MRC	15007^WBC Histogram. Middle Line Adjusted^99MRC
WBC Histogram Bitmap Data	ED	15008	WBC Histogram. BMP	99MRC	15008^WBC Histogram. BMP^99MRC
RBC Histogram Binary Data	ED	15050	RBC Histogram. Binary	99MRC	15050^RBC Histogram. Binary^99MRC
RBC Histogram Left Discriminator	NM	15051	RBC Histogram. Left Line	99MRC	15051^RBC Histogram. Left Line^99MRC
RBC Histogram Right Discriminator	NM	15052	RBC Histogram. Right Line	99MRC	15052^RBC Histogram. Right Line^99MRC
RBC Histogram Original Data Length	NM	15053	RBC Histogram. Binary Meta Length	99MRC	15053^RBC Histogram. Binary Meta Length^99MRC
RBC Histogram Left Discriminator Adjusted Mark	IS	15054	RBC Histogram. Left Line Adjusted	99MRC	15054^RBC Histogram. Left Line Adjusted^99MRC
RBC Histogram Right Discriminator Adjusted Mark	IS	15055	RBC Histogram. Right Line Adjusted	99MRC	15055^RBC Histogram. Right Line Adjusted^99MRC
RBC Histogram Bitmap Data	ED	15056	RBC Histogram. BMP	99MRC	15056^RBC Histogram. BMP^99MRC
PLT Histogram Binary Data	ED	15100	PLT Histogram. Binary	99MRC	15100^PLT Histogram. Binary^99MRC
PLT Histogram Left Discriminator	NM	15111	PLT Histogram. Left Line	99MRC	15111^PLT Histogram. Left Line^99MRC
PLT Histogram	NM	15112	PLT Histogram.	99MRC	15112^PLT

Right Discriminator			Right Line		Histogram. Right Line^99MRC
PLT Histogram Original Data Length	NM	15113	PLT Histogram. Binary Meta Length	99MRC	15113^PLT Histogram. Binary Meta Length^99MRC
PLT Histogram Left Discriminator Adjusted Mark	IS	15114	PLT Histogram. Left Line Adjusted	99MRC	15114^PLT Histogram. Left Line Adjusted^99MRC
PLT Histogram Right Discriminator Adjusted Mark	IS	15115	PLT Histogram. Right Line Adjusted	99MRC	15115^PLT Histogram. Right Line Adjusted^99MRC
PLT Histogram Bitmap Data	ED	15116	PLT Histogram. BMP	99MRC	15116^PLT Histogram. BMP^99MRC
DIFF Scattergram Bitmap Data	ED	15200	WBC DIFF Scattergram. BMP	99MRC	15200^WBC DIFF Scattergram. BMP^99MRC
Abnormal alarm information					
WBC Abn scattergram	IS	12000	WBC Abnormal scattergram	99MRC	12000^WBC Abnormal scattergram^99MRC
Abnormal WBC Histogram	IS	12001	WBC Abnormal histogram	99MRC	12001^WBC Abnormal histogram^99MRC
Leucocytosis	IS	12002	Leucocytosis	99MRC	12002^Leucocytosis^99MRC
Leucopenia	IS	12003	Leucopenia	99MRC	12003^Leucopenia^99MRC
Neutrophilia	IS	12004	Neutrophilia	99MRC	12004^Neutrophilia^99MRC
Neutropenia	IS	12005	Neutropenia	99MRC	12005^Neutropenia^99MRC
Lymphocytosis	IS	12006	Lymphocytosis	99MRC	12006^Lymphocytosis^99MRC
Lymphopenia	IS	12007	Lymphopenia	99MRC	12007^Lymphopenia^99MRC
Monocytosis	IS	12008	Monocytosis	99MRC	12008^Monocytosis^

					99MRC
Eosinophilia	IS	12009	Eosinophilia	99MRC	12009^Eosinophilia^99MRC
Basophilia	IS	12010	Basophilia	99MRC	12010^Basophilia^99MRC
WBC abnormal	IS	12011	WBC Abnormal	99MRC	12011^WBC Abnormal^99MRC
Left Shift?	IS	17790-7	WBC Left Shift?	LN	17790-7^WBC Left Shift?^LN
Immature Granulocyte?	IS	34165-1	Imm Granulocytes?	LN	34165-1^Imm Granulocytes?^LN
Abnormal/Atypical Lymphocyte?	IS	15192-8	Atypical Lymphs?	LN	15192-8^Atypical Lymphs?^LN
RBC Lyse Resist?	IS	34525-6	rstRBC	LN	34525-6^rstRBC^LN
Erythrocytosis	IS	12012	Erythrocytosis	99MRC	12012^Erythrocytosis^99MRC
RBC abnormal distribution	IS	12013	RBC Abnormal distribution	99MRC	12013^RBC Abnormal distribution^99MRC
Anisocytosis	IS	15150-6	Anisocytosis	LN	15150-6^Anisocytosis^LN
Macrocytosis	IS	15198-5	Macrocytes	LN	15198-5^Macrocytes^LN
Microcytosis	IS	15199-3	Microcytes	LN	15199-3^Microcytes^LN
Dimorphologic	IS	10379-6	RBC Dual Pop	LN	10379-6^RBC Dual Pop^LN
Anemia	IS	12014	Anemia	99MRC	12014^Anemia^99MRC
Hypochromia	IS	15180-3	Hypochromia	LN	15180-3^Hypochromia^LN
HGB Abn/Interfere?	IS	12015	HGB Interfere	99MRC	12015^HGB Interfere^99MRC
PLT Abnormal Distribution	IS	12016	PLT Abnormal Distribution	99MRC	12016^PLT Abnormal Distribution^99MRC
Thrombocytosis	IS	12017	Thrombocytosis	99MRC	12017^Thrombocytosis^99MRC
Thrombopenia	IS	12018	Thrombopenia	99MRC	12018^Thrombopenia

					^99MRC
PLT Clump?	IS	7796-6	Platelet Clump?	LN	7796-6^Platelet Clump?^LN

Table 11 Units of communication data

Parameters' units displayed on the screen of the BC-5380	Units of communication data (OBX-6)
10^12/L	10*12/L
10^9/L	10*9/L
10^6/uL	10*6/uL
10^4/uL	10*4/uL
10^3/uL	10*3/uL
10^2/uL	10*2/uL
mL/L	mL/L
/nL	/nL
/pL	/pL
g/L	g/L
g/dL	g/dL
L/L	L/L
mmol/L	mmol/L
%	%
fL	fL
um^3	um3
pg	pg
fmol	fmol
amol	amol
Year (age unit)	yr
Month (age unit)	mo
Day (age unit)	d
Hour (age unit)	hr

3. Part of the OBX messages adopt the customized enumeration values listed in the following table.

Data item	Enumeration values
Take Mode	The values are the following enumerations: "O" – open vial "A" – autoloading "C" – closed vial

Blood Mode	The values are the following enumerations: “W”- whole blood “P” – prediluted
Test Mode	The values are the following enumerations: “CBC” “CBC+DIFF”
Age	The values are the numeric data and the units are the following enumerations: “yr” - year “mo” – month “d” - day “hr” - hour
Blood Type ABO	The values are the following enumerations: “A” “B” “O” “AB”
Blood Type RH	The values are the following enumerations: “RH+” “RH-”
Qc Level	The values are the following enumerations: “L”- low “M”- normal “H”- high
Adjusting marks of histogram discriminators and flags	OBX-2 data type is “IS”; the values are the following enumerations: “T”- true “F”- false

4. Histogram data: according to the software setup, there are several cases for the communication of the histogram data.

(1)Do not transmit the histogram data.

(2)Transmit histogram data in the form of bitmap. In the OBX segment, the value of the data type field is “ED”; the value of data is in the form of “^Image^BMP^Base64^……histogram data in the form of bitmap……”, the “image” herein indicates the image data is transmitted; the “BMP” is the customized sub-data type, it indicates the BMP bitmap is transmitted; “Base64” indicates the coding method of the data.

(3)Transmit binary histogram data. In the OBX segment, the value of the data type field is “ED”; the value of data is in the form of “^Application^Octet-stream^Base64^……histogram data……”; “Application^Octer-stream” herein is the sub-data of HL7 standard, indicating the

binary data type defined by the application program; “Base64” indicates the coding method of the data.

Note: to transmit the histogram data in the form of bitmap or binary is determined by the ID field in the OBX segment.

5. Scattergram data: when transmitting bitmap data, in the OBX segment, the value of the data type field is “ED”; the value of data is in the form of “^Image^BMP^Base64^……scattergram data in the form of bitmap……”. The “Image^BMP^Base64” indicates the bitmap data is of BMP type and coded by Base64.

6. Age communication: the age in the patient information will be transferred as an OBX message segment composed of an integer and the age unit. If the age in the IPU software is displayed as “<1”, then the age value in the communication is “0”.

C.6 Appendix: Base64 coding procedures

(1) Select the 3 adjacent bytes (i.e. 24 bit) from the data stream to be coded; from left to right, divide them into 4 groups of 6-bit; then, ASCII string is obtained by mapping as per the Table 12.

Initial data	15H	A3H		4BH
Binary data	00010101	10100011		01001011
6-bit group obtained after dividing	000101	011010	001101	001011
Corresponding coding value	5H	1AH	0DH	0BH
Corresponding character	F	a	N	L

Table 12 Base64 mapping

Value/Code	Value/Code	Value/Code	Value/Code
0 A	17 R	34 I	51 z
1 B	18 S	35 j	52 0
2 C	19 T	36 k	53 1
3 D	20 U	37 l	54 2
4 E	21 V	38 m	55 3
5 F	22 W	39 n	56 4
6 G	23 X	40 o	57 5
7 H	24 Y	41 p	58 6
8 I	25 Z	42 q	59 7
9 J	26 a	43 r	60 8
10 K	27 b	44 s	61 9

11 L	28 c	45 t	62 +
12 M	29 d	46 u	63 /
13 N	30 e	47 v	
14 O	31 f	48 w	(pad) =
15 P	32 g	49 x	
16 Q	33 h	50 y	

(2) Repeat the coding of procedure (1) continuously till finish coding the data stream.

When the data left is less than 3 bytes, 0 is used to complement to the right. If the whole 6-bit group obtained is composed of 0, then it is mapped to the “=” character. When one byte is left, then the obtained coding string consists of two “=” characters; when two bytes are left, then the obtained coding string consists of one “=” character. The two cases are demonstrated below:

① Initial data 0AH

00001010

Data obtained after complementing	00001010	00000000	00000000	
6-bit groups obtained after dividing	000010	100000	000000	000000
Corresponding values	02H	20H	00H	00H
Corresponding characters	C	g	=	=

② Initial data 0AH 0BH

00001010 00001011

Data obtained after complementing	00001010	00001011	00000000	
6-bit groups obtained after dividing	000010	100000	101100	000000
Corresponding values	02H	20H	2CH	00H
Corresponding characters	C	g	s	=

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